

Akshat Sharma

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EXPERIENCE

Software Developer

Apr 2024 – Present

Tessonics

Windsor, ON

- Built a **C++** PLC simulator that mirrors real production traffic, accelerating integration tests for the **RIWA** platform.
- Engineered a real-time Hall-effect current-detection algorithm in **C++**, achieving <10 ms end-to-end latency.
- Authored backward-compatible 32-bit **C++** DLLs that exposes a drop-in API, bridging legacy software to a modern **C++** inference server via WebSocket and delivering real-time deep-learning predictions with no code rewrites.
- Built a **Python**-based **Debian** packaging pipeline and self-hosted **GitHub Actions** runner, fully automating CI/CD and streamlining RIWA software deployment.
- Contributed to the design and development of **PostgreSQL** database schemas from scratch for the RIWA platform, creating migration scripts and tooling to manage schema evolution across production environments.
- Developed **Go**-based API mocking infrastructure to enable **React** web app development and load testing without hardware dependencies, accelerating frontend iteration with simulated **WebSocket** data streams.
- Automated cleaning, analysis, and visualization of **EEG** & Hypnogram time-series data using **NumPy** and **matplotlib**.
- Developed **STM32** firmware (**C**) that integrates the **MCP4725** DAC and **RX8900** RTC for the SONYA sleep wearable.
- Implemented a **Bluetooth Low Energy (BLE)** GATT service on the STM32 that lets users remotely adjust stimulation intensity and toggle operating modes on the sleep-wearable in real time.
- Optimized **ISM330DHCX** IMU data capture via FIFO and I²C, ensuring synchronized, real-time motion streams for onboard processing.

TECHNICAL SKILLS

Languages: C | C++17 | C++20 | Python | Java | Kotlin | Go | JavaScript

Data Science & ML: TensorFlow | PyTorch | Keras | OpenCV | NumPy | SciKit-Learn | matplotlib

Frameworks & Tools: React | Git | Docker | GitHub Actions | CMake | Conan

Database: PostgreSQL | SQLite | MySQL | Redis

Embedded / HW: STM32CubeIDE | FreeRTOS | I²C | SPI | Arduino

PROJECTS

Charty | Rust, WebSocket, Finnhub API

Jan 2026 - Present

- Built a terminal-based stock market analysis application in **Rust** featuring interactive candlestick charts and real-time price streaming via **WebSocket**.
- Implemented multiple view modes including historical charts with configurable timeframes and live ticker/candle aggregation for real-time market data.
- Designed an event-driven TUI architecture handling concurrent data fetching, WebSocket streaming, and responsive keyboard navigation.

EDUCATION

University of Windsor

Jan 2021 – Apr 2024

Bachelor of Computer Science Honours

Windsor, ON

- *Specialization in Artificial Intelligence, Minor in Mathematics*

VOLUNTEER & AWARDS

• Event Coordinator, Computer Science Society

2023 – 2024

Organized events that served 100+ students each term and streamlined logistics to improve attendee experience.

• Dean's Honour Roll

2021 – 2024

University of Windsor